

1 (In)determinacy in woody perennials: limits and opportunities
2 for timing growth in a changing climate

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Abstract

When and how much organisms grow given environmental constraints are fundamental questions in biology, and especially pressing in the context of climate change as we strive to predict biomass production and carbon sequestration. While environmental factors such as temperature and water availability directly regulate plant growth, the timing and rate of growth are also governed by a plant's phenological sequence—its genetically programmed developmental stages. This sequence is critical for predicting climate responses but is often underappreciated.

Here, we leverage the concept of (in-)determinacy in growth and development—which captures how flexibly plants produce organs from preformed cells versus initiating them *de novo*—to propose a new framework for predicting tree growth responses to climate change.

We propose that: 1) determinate growth is an adaptation to strongly seasonal climates, restricting tissue investment to a short favorable period before dormancy; 2) indeterminate species are more exposed to increasing climatic extremes (spring/fall frost, drought) but may recover more effectively from resulting damage; and 3) species with greater indeterminacy are more likely to benefit from lengthening frost-free growing seasons. Thus, future carbon sequestration may depend not only on environmental changes, but also on the degree of determinacy set by intrinsic genetic programming.

Introduction

Investing the right amount of resources in growth at the right time is of crucial importance to the survival and fitness of any living organism. The topic of growth strategies and habits has a long history in science, spanning the fields of genetics, genomics, physiology and ecology across the animal and plant kingdoms (Stearns, 1998). At its core lies the concept of determinacy—the classification of organisms as either reaching a fixed size at maturity or continuing to grow throughout their lives. Like mollusks, fish and reptiles, plants add biomass to their primary bodies as long as they live and are therefore considered ‘indeterminate growers’ (Ejzmond *et al.*, 2010).

Various terms have emerged to describe this phenomenon across spatial and temporal scales, e.g. from a single cell to the whole organism, and from a single season to an entire lifetime (McDaniel, 1992; Karkach, 2006). In plants, organs can develop from preformed embryonic tissue (e.g. primordia in overwintering buds), form *de novo*, or both, depending on species and life form. While these underlying developmental and meristematic constraints apply broadly across the plant kingdom, they play a particularly central role in the life cycle of woody perennials, especially trees, which must balance growth, survival, and reproduction over long lifespans in highly seasonal environments. As we use it here, (in)determinacy in woody perennials refers to the ability to:

- a) preform organs in one year as a future investment, which are deployed rapidly in the following spring with no further primary shoot growth during that season (determinate strategy)
- b) maintain a continuous or episodic meristem activity by forming new organs that mature during the current growing season (indeterminate strategy).

In this perspective we revisit classical concepts of determinacy and growth patterns of the shoot apical meristem in woody perennials, generalize them across meristem types and position them within the context of climate change. Although we focus here on woody perennials, the concept of (in)determinacy is broadly applicable across vascular plants, including herbaceous species; trees are used as a model system in which the ecological and climatic consequences of determinacy are particularly pronounced. To provide context, we briefly introduce the fundamental limitations to plant growth in extra-tropical regions by linking external environmental drivers and internal growth control mechanisms.

Seasonality of temperature, soil moisture and light

The further one travels from the equator towards the poles, the more strongly plants are confined to a shrinking time window of opportunity for growth set primarily by low temperatures. Below c. 5 °C, metabolic activity slows to an extent where growth and development largely cease (Schenker *et al.*, 2014; Rossi *et al.*, 2008; Körner, 2008). More importantly, sub-freezing temperatures can cause severe damage to plant tissue if they occur at vulnerable developmental stages, e.g. during or after leaf unfolding, prior to fruit maturation, or before cold acclimation in fall (Sakai & Larcher, 1987; Baumgarten *et al.*, 2023). While annual plant species accommodate their entire life cycle within this window, perennial (woody) species are forced to partition their growing phase seasonally, with periods of activity alternating with periods of dormancy (Delpierre *et al.*, 2016). This is referred to as intermittent or rhythmic (as opposed to continuous) growth.

High temperatures can directly inhibit growth by exceeding species-specific physiological thresholds (O’Sullivan *et al.*, 2017). In addition, increased evaporative demand and reduced precipitation can lower soil moisture, leading to water stress that slows or stalls growth and, in extreme cases, causes leaf scorching or premature senescence (Hsiao, 1973; Pugnaire *et al.*, 1999; Etzold *et al.*, 2021; Estiarte & Peñuelas, 2015). Together, these temperature and soil moisture limitations act as environmental filters, narrowing the period during which growth is possible (Figure 1).

Light also influences plant growth through two key processes. First, light intensity provides the primary energy source that drives photosynthesis and, consequently, source activity. Second, light carries a wealth of information through its quality and the length of daily light and dark periods (photoperiod), which mediate many physiological processes, including bud set and dormancy transitions (Wang *et al.*, 2024). Both factors may contribute to the common observation that tree growth often peaks near the summer solstice in extra-tropical regions (Rossi *et al.*, 2006; Etzold *et al.*, 2021; Luo *et al.*, 2018) or that a window of temperature sensitivity ‘opens’ around the summer solstice to synchronize reproductive efforts over very large spatial scales (Journé *et al.*, 2024).

Internal programming of plants

Given our relatively robust understanding of environmental influences on plant growth, it may seem straightforward to predict how plants will grow under current and future climatic conditions. Yet,

this has proven challenging, particularly when predicting responses under scenarios with extended, climatically favorable growing seasons (Zohner *et al.*, 2021). Here we propose that accurate predictions require incorporating another key, but often overlooked, factor: internal growth control—the genetically encoded developmental program that may prevent further growth *despite* favorable environmental conditions.

While plants have evolved numerous morphological adaptations to tolerate or avoid harmful environmental conditions, most temperate and boreal woody perennials cope with seasonal fluctuations in temperature and moisture by temporally escaping these conditions. They do this by aligning their life-history events (phenology) with a cycle of growth and dormancy that balances survival, reproduction, and growth over the long lifespan typical of many tree species (Delpierre *et al.*, 2016). Hence, the phenological sequence can impose abrupt internal switches in resource allocation—from vegetative growth to reproduction (flowering, fruit maturation) and storage (Stearns, 1989; Chapin *et al.*, 1990). These transitions act as additional internal filters, further narrowing the effective window during which vegetative growth can occur, regardless of environmental potential (Figure 1).

The concept of (in)determinate growth

In annual plants, growth ends with the production of flowers to form fruits and seeds: a signal in the apical meristem triggers a sudden switch in resource investment from vegetative growth to building a reproductive structure with no point of return, signaling the end of the life-cycle (Poethig, 2003; Huijser & Schmid, 2011). A shoot axis that terminates in a flower or other modified structure and ceases meristematic activity is considered ‘determinate’ (Barthélémy & Caraglio, 2007). In woody perennial species, however, flowers are typically produced on lateral shoots or buds, preserving the main vegetative axis and allowing continued structural expansion. However, ‘determinate growth’ also refers to a temporal suspension of the meristem, which leads to a time lag between the formation of organs and their expansion (Kozlowski, 2012; Hallé *et al.*, 1978).

In their first year, all seedlings exhibit indeterminate growth. In subsequent years, many temperate and boreal tree species preform their new shoot increments during the previous growing season, overwintering them as primordial shoots with leaves and internodes in hardened buds to be ‘ready-to-go’ when spring arrives. Once the leaves flush and the preformed internodes elongate, species differ in

133 how long the shoot apical meristem remains competent to continue organ production (Figure 2). Some
 134 species exhibit strongly determinate shoot extension, in which the terminal meristem is intrinsically
 135 downregulated soon after the spring flush and remains inactive for the rest of the season, consistent
 136 with a prolonged internal suppression (paradormancy; Lang *et al.*, 1987; Figure 4; e.g. *Fraxinus* spp.,
 137 *Acer* spp. and *Prunus* spp.). At the other extreme, highly indeterminate species maintain an active
 138 shoot apical meristem and can continue producing new leaves and internodes on top of the preformed
 139 shoot as long as conditions remain favorable, sometimes stretching their indeterminate growth well into
 140 autumn until low temperature eventually forces them to stop (e.g. *Sequoia* spp., *Cupressus* spp. and
 141 *Alnus* spp.). Between these extremes, many species occupy an intermediate position: they continue
 142 to produce neoformed leaves and internodes after the preformed increment, but growth cessation is
 143 often coordinated by seasonal cues—frequently photoperiod acting through hormonal regulation (e.g.
 144 *Pinus* spp., *Betula* spp. and *Populus* spp.; Maurya & Bhalerao, 2017; Triozzi *et al.*, 2018; Böhlenius
 145 *et al.*, 2006). Indeterminate growth can occur either through continuous activity of the shoot apical
 146 meristem forming new leaves and internodes without buds (free growth), or through the formation
 147 and subsequent flushing of buds mid-season without a period of dormancy (second flushing or lammas
 148 growth; Figure 2).

149
 150 Although this concept is often presented dichotomously (Kozłowski & Pallardy, 1997; Lechowicz, 1984),
 151 but see Kikuzawa, 1983; Damascos *et al.*, 2005), with species classified as either determinate or in-
 152 determinate growers, most species likely exist along a gradient with numerous intermediate forms.
 153 For instance, many oaks (*Quercus* spp.) are considered determinate growers, but can exhibit multiple
 154 flushes. Similarly, Douglas-fir (*Pseudotsuga menziesii*) gradually adopts a more determinate growth
 155 habit with maturity (Borchert, 1976; Heuret *et al.*, 2006). However, a gradual quantification of the
 156 degree of determinacy on the species level is possible at least for the shoot apical meristem (Box 1).
 157 See Figure 4 for a literature-based classification of determinacy of some common species with many
 158 among them falling into an intermediate position.

159

160 Evolutionary trade-offs: the cost of certainty and flexibility

161 In most temperate and boreal ecosystems, determinate and indeterminate growth strategies co-occur,
 162 indicating that neither strategy is universally superior. Instead, they reflect contrasting solutions to

163 a fundamental trade-off between growth certainty and growth flexibility under seasonal and variable
164 environments.

165

166 Indeterminate species are characterized by continued or episodic activity of the shoot apical meristem
167 during the growing season. This does not necessarily imply faster intrinsic growth rates, but rather a
168 greater capacity to extend growth in time when conditions remain favorable. Such temporal flexibility
169 allows individuals to exploit transient resource availability and recover from partial damage, traits
170 that are advantageous in disturbed or early-successional environments. Accordingly, indeterminate
171 growth is frequently associated with early successional species (Marks, 1975). However, maintaining
172 active meristems later into the season also increases exposure to environmental stressors such as sum-
173 mer drought and early autumn frost, thereby raising the risk of tissue loss or incomplete maturation
174 (Figure 4). As a result, indeterminate growth strategies often involve higher interannual variability
175 in performance and, in some cases, shorter tissue or organismal lifespans (Millet *et al.*, 1999; Brien
176 *et al.*, 2020).

177

178 In contrast, determinate species preform a substantial portion of their shoot growth in advance and
179 deploy this tissue during a restricted period of the growing season. This strategy limits short-term
180 plasticity, as growth is largely constrained by conditions experienced during bud formation in the
181 previous year. At the same time, it reduces exposure of sensitive tissues to late-season stress and
182 synchronizes growth with the most predictable and favorable part of the season. As a consequence,
183 determinate growth strategies are often associated with stronger local adaptation and reduced sensi-
184 tivity to short-term climatic fluctuations (Leites & Benito Garzón, 2023).

185

186 These contrasting strategies help explain the coexistence of determinate and indeterminate species
187 within the same landscapes. For example, western larch (*Larix occidentalis*) and Douglas-fir (*Pseudot-
188 suga menziesii*) frequently co-occur despite marked differences in growth determinacy. Western larch
189 exhibits more flexible, indeterminate shoot growth and performs well in environments shaped by fre-
190 quent disturbance, whereas Douglas-fir shows more determinate growth and stronger local adaptation,
191 conferring stability under predictable seasonal regimes (Roskilly & Aitken, 2024). Such differences
192 suggest niche differentiation along gradients of disturbance frequency, environmental stress, and cli-
193 matic predictability, rather than outcomes driven by straightforward competitive dominance.

194

195 Together, these patterns indicate that growth determinacy reflects a trade-off between flexibility and
 196 reliability in seasonal environments. Under ongoing climate change—marked by longer growing seasons,
 197 greater climatic variability, and increasing environmental stress in many regions—it remains uncertain
 198 whether determinate or indeterminate growth strategies will be favored. Resolving both how trees
 199 regulate their (in)determinacy and which levels along this trait continuum are advantageous under
 200 future conditions will be critical for predicting shifts in species performance, coexistence, and forest
 201 composition.

202 The performance of (in)determinacy with climate change

203 Spring warming has advanced the onset of leaf emergence by up to a month compared to pre-industrial
 204 times (Vitasse *et al.*, 2022), reflecting an earlier reactivation of shoot apical and cambial meristems
 205 under warmer conditions. In contrast, the cessation of growth in autumn has shifted far less than
 206 expected given the extension of climatically favorable conditions (Zani *et al.*, 2020; Zohner *et al.*,
 207 2023). As a result, phenological sequences are increasingly observed to shift as a whole toward spring
 208 rather than being symmetrically extended at both ends of the growing season (Keenan & Richardson,
 209 2015; Meng *et al.*, 2024). This asymmetric response highlights a key constraint relevant to growth
 210 (in)determinacy: while growth onset is highly plastic and environmentally responsive, the duration of
 211 meristematic activity later in the season appears more tightly regulated and differs markedly among
 212 species. Such constraints on growth cessation help explain why longer growing seasons do not neces-
 213 sarily translate into increased growth or biomass production (Zani *et al.*, 2020; Wolkovich *et al.*, 2025).

214

215 Along the continuum of growth determinacy, we hypothesize that species positioned toward the more
 216 determinate end will largely maintain their fixed seasonal growth schedules under climate warming,
 217 with limited capacity to extend growth later into the season. In regions where climate change primarily
 218 increases heat and water stress during mid to late summer, such species are therefore expected to show
 219 little change in annual biomass production, as their restricted growth window increasingly overlaps
 220 with unfavorable conditions (Figure 3).

221 In contrast, we predict that species toward the indeterminate end of the continuum will be more ca-
 222 pable of prolonging meristematic activity when conditions remain favorable, potentially resulting in
 223 higher annual production under some future climate scenarios (Caffarra & Donnelly, 2011; Richardson

224 *et al.*, 2009; Zohner *et al.*, 2020, Figure 3). This capacity for temporal extension of growth is partic-
225 ularly common among early successional species, which tend to leaf out earlier in spring and senesce
226 later in autumn than late-successional species. Their earlier leaf-out is consistent with generally lower
227 chilling and forcing requirements for bud burst and dormancy release (Heide, 1993; Laube *et al.*, 2014;
228 Hu *et al.*, 2022; Baumgarten *et al.*, 2021)

229

230 Extreme climatic events and increasing environmental stress provide an important context for evalu-
231 ating these predictions. More determinate growth strategies concentrate growth earlier in the season
232 and may therefore escape unfavorable growing conditions by operating within a relatively narrow
233 window of potential growth. Under historical climatic regimes, this strategy provided ample safety
234 margins against late-season stress. Where climate change introduces novel or intensified drought stress
235 within this window, however, determinate species are likely to experience reduced annual production
236 and limited capacity for recovery within the same growing season once canopy development has been
237 compromised (Silvestro *et al.*, 2023, Figure 3).

238 Species with more indeterminate growth habits often remain green and physiologically active for longer
239 periods, leafing out early in spring and senescing late in autumn, with deciduous species occasionally
240 retaining functional foliage until the first freezing events (Figure 3). As a consequence, a substantial
241 fraction of their growth period coincides with parts of the growing season during which drought stress
242 is projected to intensify under future climates, particularly in water-limited regions (Figure 3). While
243 this extended activity increases exposure to environmental extremes, it also creates opportunities for
244 compensatory growth when conditions remain favorable.

245 In essence, the flexible growth schedule of indeterminately growing species may allow plants to: (1)
246 produce tissue better adapted to harsh environmental conditions as it is formed in the current season
247 (e.g., smaller leaves with higher specific leaf area), and (2) catch up and compensate later in the season
248 by another production boost (Stevens & Lindroth, 2005). Mediterranean ecosystems provide a clear
249 example of this flexibility, where many species suspend growth during summer drought but resume
250 meristematic activity in autumn or winter when conditions allow (Volaire *et al.*, 2023). Evidence that
251 trees can take advantage of such a ‘second growing season’ after summer drought has been found in
252 pines in Mediterranean climates through polycyclic flushing—a form of indeterminate growth (Girard
253 *et al.*, 2011, Figure 3).

254

We therefore hypothesize that climate change will increasingly disrupt established phenological cycles, favoring species and genotypes that are more plastic in rearranging their activities by resuming growth and/or restoring reserves (e.g., starch, nutrients) later in the year, thereby recovering from and compensating for stress-induced damage and losses. Such differential capacity to exploit newly available growth opportunities or to recover after stress is expected to intensify competition among co-occurring species by altering the timing and magnitude of resource uptake, potentially leading to the reassembly of forest communities toward species occupying more indeterminate positions along the growth continuum.

(In)determinacy beyond leaf tissue

Although (in)determinacy is most clearly defined *sensu stricto* at the shoot apical meristem, its relevance extends beyond primary shoot growth because shoot extension imposes fundamental structural constraints on whole-plant growth. Primary growth governs crown expansion and annual leaf deployment, thereby shaping total leaf area and canopy photosynthetic capacity—key determinants of biomass production that have often been underappreciated in studies of tree growth (Girard *et al.*, 2011; Soolanayakanahally *et al.*, 2015). At the same time, primary shoot growth establishes the vascular framework within which secondary growth occurs, such that the size and extent of a shoot increment determine the length and surface area over which cambial activity can later be expressed. Empirical studies report close coordination between primary shoot growth and stem or radial growth within seasons, indicating that variation in shoot extension is tightly linked to variation in wood formation (Cuny *et al.*, 2012; Zhang *et al.*, 2019).

Despite its central role in tree development, primary shoot growth remains far less frequently measured than radial growth, an imbalance that is striking given that shoot extension occurs at the decimeter scale, while radial growth—expressed at the micrometer scale—can be monitored continuously and with relative ease using dendrometers or microcoring (see Box 1 for alternative approaches to quantify primary growth).

In the vascular cambium, seasonal patterns of cell division and differentiation are temporally structured but cannot be interpreted in terms of preformation or neoformation. Cambial initial cells divide to produce daughter cells that mature into functional xylem and phloem (Valdovinos-Ayala *et al.*, 2022). The timing and duration of cambial cell division constrain the rate and seasonal extent of

285 secondary growth by limiting the number of cells entering differentiation, while subsequent cell matu-
 286 ration imposes additional carbon and resource demands that can further restrict wood formation (Lupi
 287 *et al.*, 2010). At the same time, cambial cell production can cease even when environmental conditions
 288 remain favorable and the canopy is still photosynthetically active, pointing to internal regulation of
 289 secondary growth (Buttò *et al.*, 2020; Arend *et al.*, 2024).

290 Below-ground meristems appear to follow a more opportunistic strategy. Root apical meristems can
 291 remain active whenever local conditions permit, with rhizotron studies documenting root growth during
 292 winter under suitable soil temperatures (Lyford & Wilson, 1966), suggesting that roots often lack a
 293 clearly defined dormant phase (Radville *et al.*, 2016; Marchand *et al.*, 2025). However, the extent to
 294 which below-ground growth dynamics reflect direct environmental control, coordination with above-
 295 ground growth, or genetically constrained strategies remains unresolved (Abramoff & Finzi, 2015;
 296 Makoto *et al.*, 2020; Silvestro *et al.*, 2025; Campioli *et al.*, 2024).

297 Together, these observations highlight that while (in)determinacy is most rigorously defined at the
 298 shoot apical meristem, primary shoot growth exerts a disproportionate influence on whole-plant per-
 299 formance by simultaneously shaping canopy source capacity and the structural context for downstream
 300 sink activity. As such, determinacy at the shoot level provides a meaningful organizing principle for
 301 understanding growth regulation across plant organs, even where analogous processes operate through
 302 different mechanisms.

Future directions

A central challenge emerging from this perspective is to better understand how variation along the continuum of growth (in)determinacy shapes tree performance under changing climatic conditions. Rather than representing discrete strategies, degrees of determinacy define bounded trait spaces that differ among species but can vary with age, provenance, and environmental context. While the degree of determinacy is largely constrained at the species level, evidence from provenance trials shows meaningful intraspecific variation; for example, populations from lower latitudes often exhibit higher frequencies of second flushes than those from northern origins (Rudolph, 1964). Clarifying how these species-specific bounds and population-level adjustments interact with warming, drought, and increasing climatic variability will be critical for understanding how forest communities respond and reassemble under climate change. Extending this framework beyond woody perennials to include herbaceous species and perennial grasses may further help test the generality of (in)determinacy as a developmental constraint on growth timing, while highlighting differences arising from life form and the absence of secondary growth.

One key knowledge gap concerns how different degrees of (in)determinacy help trees avoid or buffer environmental stress while still maintaining the capacity to resume growth, repair tissues, replenish reserves, and compensate for earlier losses within the same season. Resolving the trade-off between escaping unfavorable conditions and exploiting longer or intermittently favorable growing periods is essential for anticipating how forest communities will change over the coming decades.

Understanding how universally tissues within species are formed determinately or indeterminately could provide insight into the evolution of this trait, which could be aided by cross-species analyses. Both growth forms seem to occur across most species of the same family or clade and hence appear to have evolved repeatedly in different groups (Figure 4), making it difficult to speculate on which strategy is ancestral (but see Hariharan *et al.*, 2016) and how rapidly (in)determinacy can evolve. As data on the degree of (in)determinacy across species accumulates (see Box 1), such analyses could potentially provide rapid insights into this and also aid forecasting for unsampled species.

Finally, future work should aim to link the temporal dynamics of primary (shoot and root) and secondary (cambial) meristems with patterns of reserve storage and reproductive allocation. Integrating shoot increment formation, wood production, non-structural carbohydrate dynamics, and reproduc-

334 tive effort within a common temporal framework will be essential for understanding whole-tree carbon
335 allocation. Correlating annual tree-ring formation with shoot increments—while accounting for the
336 timing of inter-annual phytomer production and distinguishing between preformed and neoformed tis-
337 sues (Figure 2)—offers a promising path toward such integration and toward improving mechanistic
338 predictions of tree growth and forest carbon dynamics under future climates.

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Box 1: Metrics and proxies related to (in)determinacy

To advance our understanding of growth strategies under climate change, we need not only improved methods for quantifying the degree of (in)determinacy—but also empirical approaches that move beyond a simple binary classification. An organ-level metric at the shoot apical meristem captures the defining features of (in)determinacy and allows it to be explicitly defined and quantified. Temporal dynamics of shoot growth provide a related but indirect proxy that can be applied from individual shoots to crowns and entire forest stands.

Ratio of preformed to neoformed organs: The ratio of the number of leaves (or leaf scars) produced during a growing season to the number of leaf primordia preformed in dormant buds provides a direct, mechanistic, and quantitative measure of shoot-level (in)determinacy. Values close to one indicate growth constrained to preformed organs, whereas values greater than one reflect the production of neoformed organs and thus indeterminate development. Preformation and neoformation have been documented for more than a century (Moore, 1909), and several studies have quantified the number of preformed primordia and compared it with the final number of emerged leaves or phytomers (Damascos *et al.*, 2005; Kikuzawa, 1983; Guédon *et al.*, 2006). Quantification typically relies on bud dissection, but emerging non-destructive approaches such as micro-CT imaging offer promising alternatives for assessing bud structure and preformation (Kovaleski *et al.*, 2019). This metric remains the most conceptually robust approach for defining and quantifying the level of (in)determinacy across species.

Seasonal growth dynamics of the shoot apical meristem: Temporal patterns of shoot elongation provide a complementary perspective on (in)determinacy at the shoot apical meristem. Across species, elongation may occur over a brief period, extend continuously through the growing season, or appear in multiple distinct growth phases, reflecting differences in how apical meristem activity is maintained or reactivated. Such dynamics can be quantified using repeated shoot measurements or phenological observations, and increasingly through high-resolution terrestrial LiDAR and photogrammetric techniques that capture shoot elongation and structural change at the scale of entire crowns or stands (Zhang *et al.*, 2019; Jin *et al.*, 2022). While temporal growth dynamics do not distinguish preformed from neoformed organs directly, they remain closely linked to shoot apical meristem activity and provide a scalable proxy for the seasonal expression of (in)determinacy. Repeated measurements also allow shoot-elongation curves to be reconstructed, whose shape differs systematically among species (single, short sigmoid vs. prolonged or multi-phase extension; Hover *et al.* (2017); Kuster *et al.* (2014)). Specifically, curve-derived parameters such as the duration of shoot expansion relative to the growing season provide a quantitative proxy for the degree of (in)determinacy.

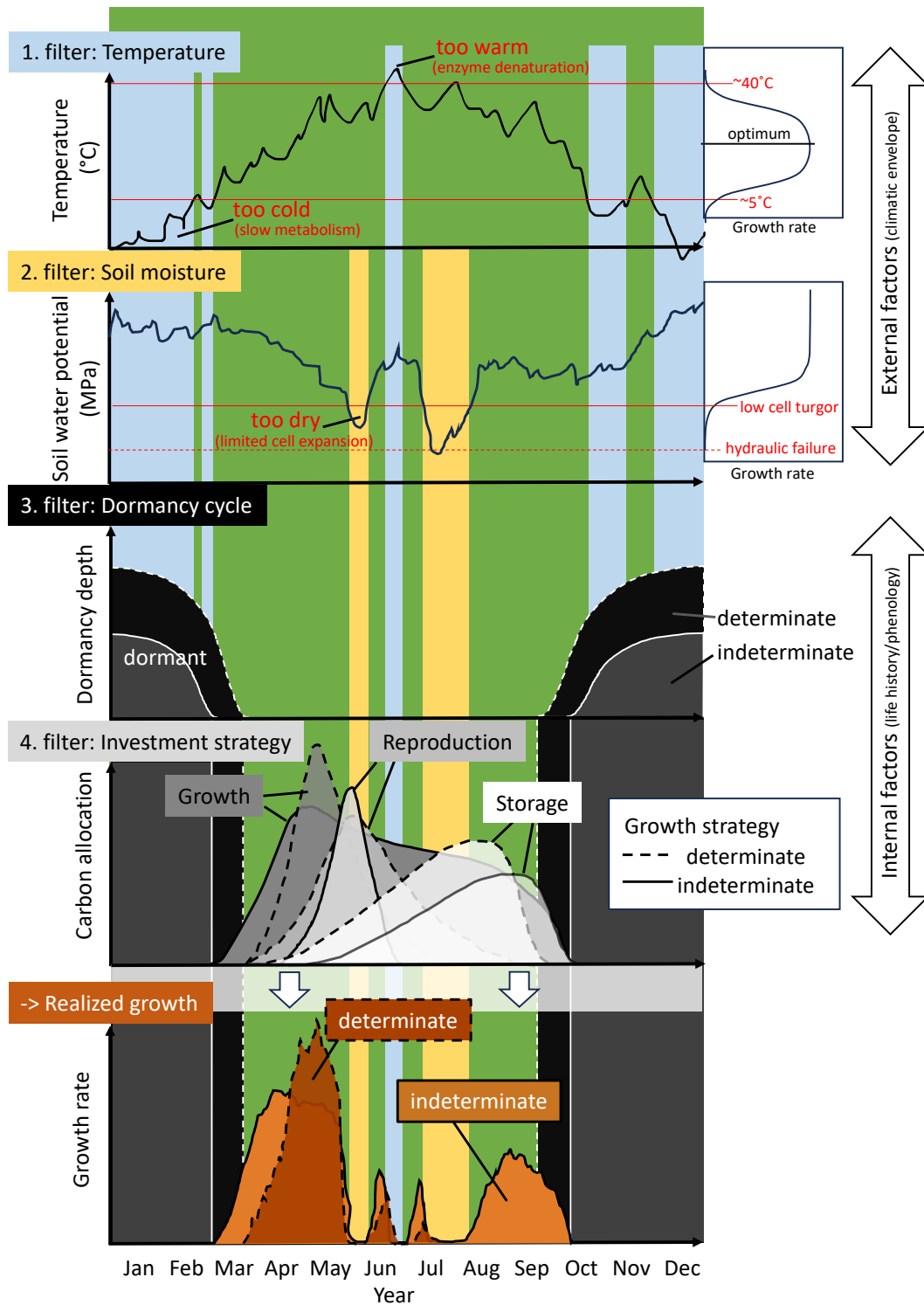


Figure 1: Schematic overview of the possible discrepancy between the potential growing season and the effectively realized vegetative growth (shoot and possibly cambial growth). When environmental factors, such as temperature and soil moisture, exceed growth-promoting thresholds they can act as filters that narrow the window of opportunity available for vegetative growth (potential light and photoperiodic constraints are not shown). The species-specific life history cycle (phenology) can impose another filter by dictating a dormancy cycle and prioritizing developmental processes other than vegetative growth (e.g. flowering, fruit maturation and storage). The degree of (in)determinacy is presented here in two extremes, although we propose to consider this trait as continuous.

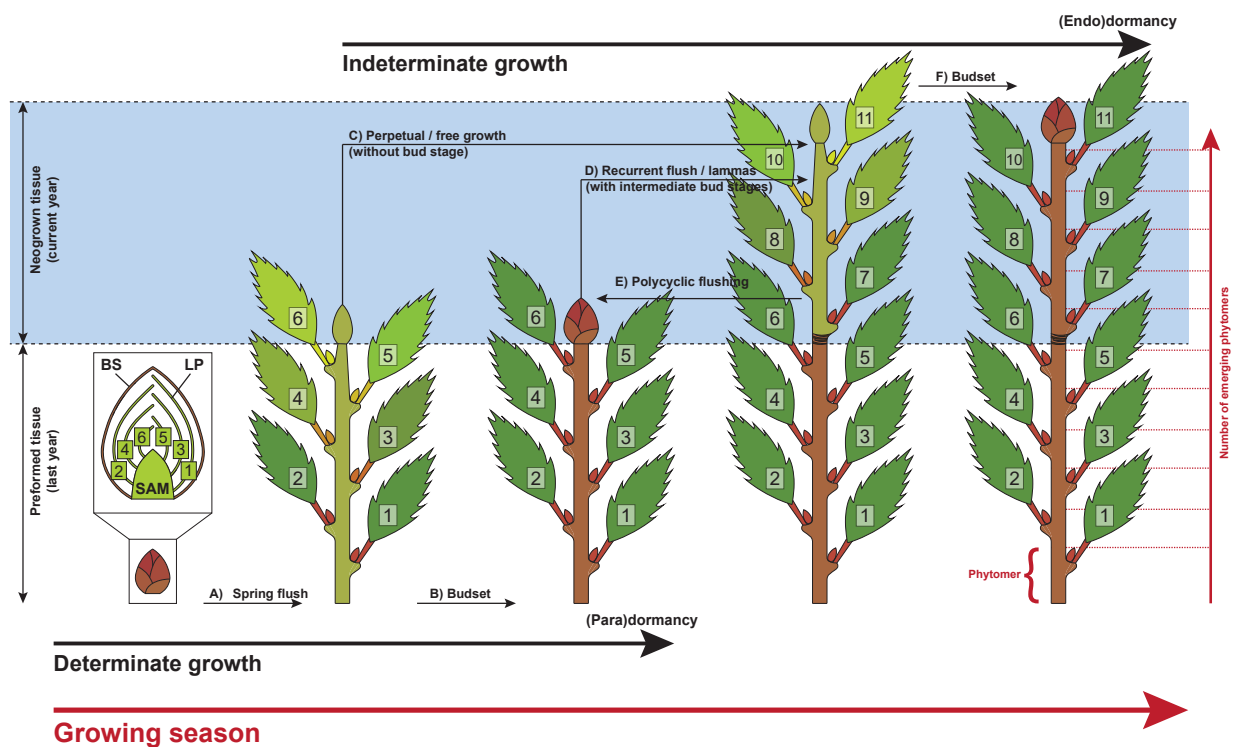


Figure 2: Determinate and indeterminate growth within one growing season for species producing terminal buds. Commonly all tree species deploy buds during their first (spring) flush from prebuilt and overwintering leaf primordia (LP) protected by bud scales (BS; A). Determinate growing species set buds (B) that are under hormonal suppression to inhibit any further activity of the shoot apical meristem – SAM (paradormancy). The basic unit of a shoot is the phytomer which is composed of a node, a leaf, the axillary bud and an internode (see bottom right). Indeterminate growing species continue to produce new phytomers directly (C) or through one (D) to several (E) intermediate bud stage(s). Finally, most species set their terminal buds (F) and enter full dormancy (endodormancy).

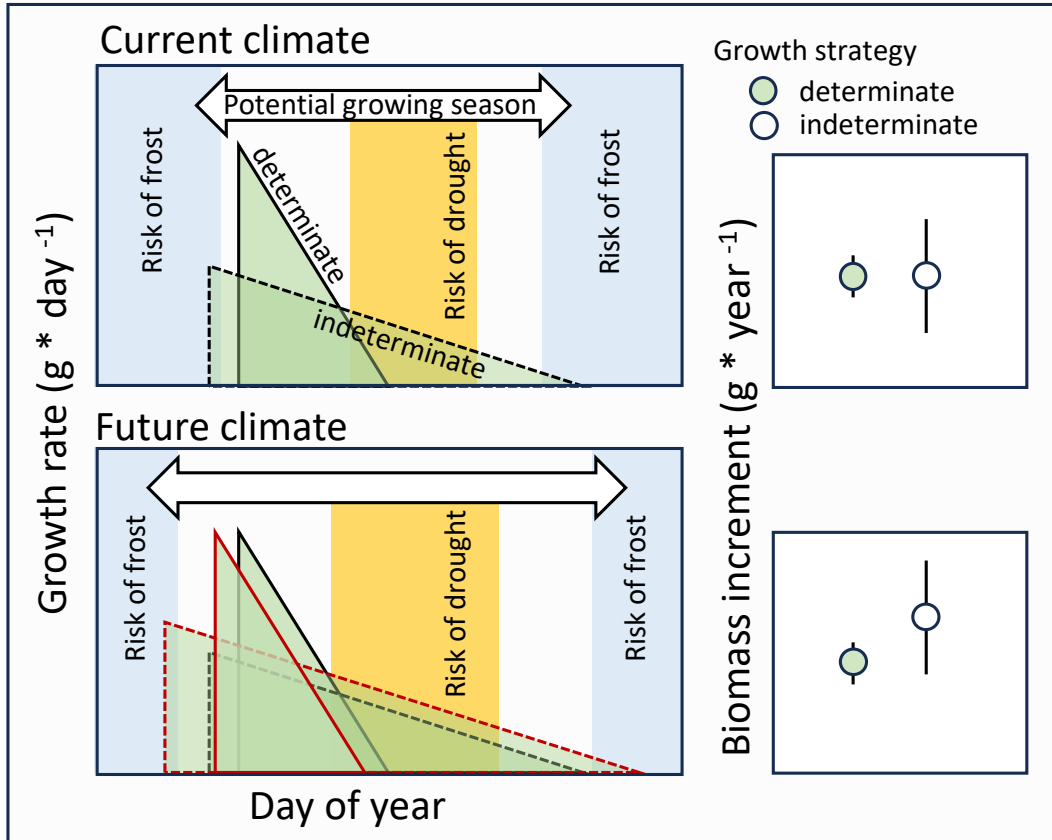


Figure 3: Hypothesized predictions of growth rates for the three major meristem (apical, cambium and root) classes of trees under current and warmer climates following an extreme determinate (A) and indeterminate (B) growth strategy. The area under the curve is summarized as yearly biomass increment in the respective bar-plot. Arrows indicate the shift of growth phenology under warmer climate conditions. Root meristems appear to be purely temperature opportunistic for both strategies, even growing during warm winter spells. The indicated genera were observed to showcase the illustrated trends. The responses of these two contrasting growth strategies might apply not only to different tree species but also within a population (e.g. along environmental gradients) and even within an individual as it transitions from the juvenile to the adult stage (ontogeny).

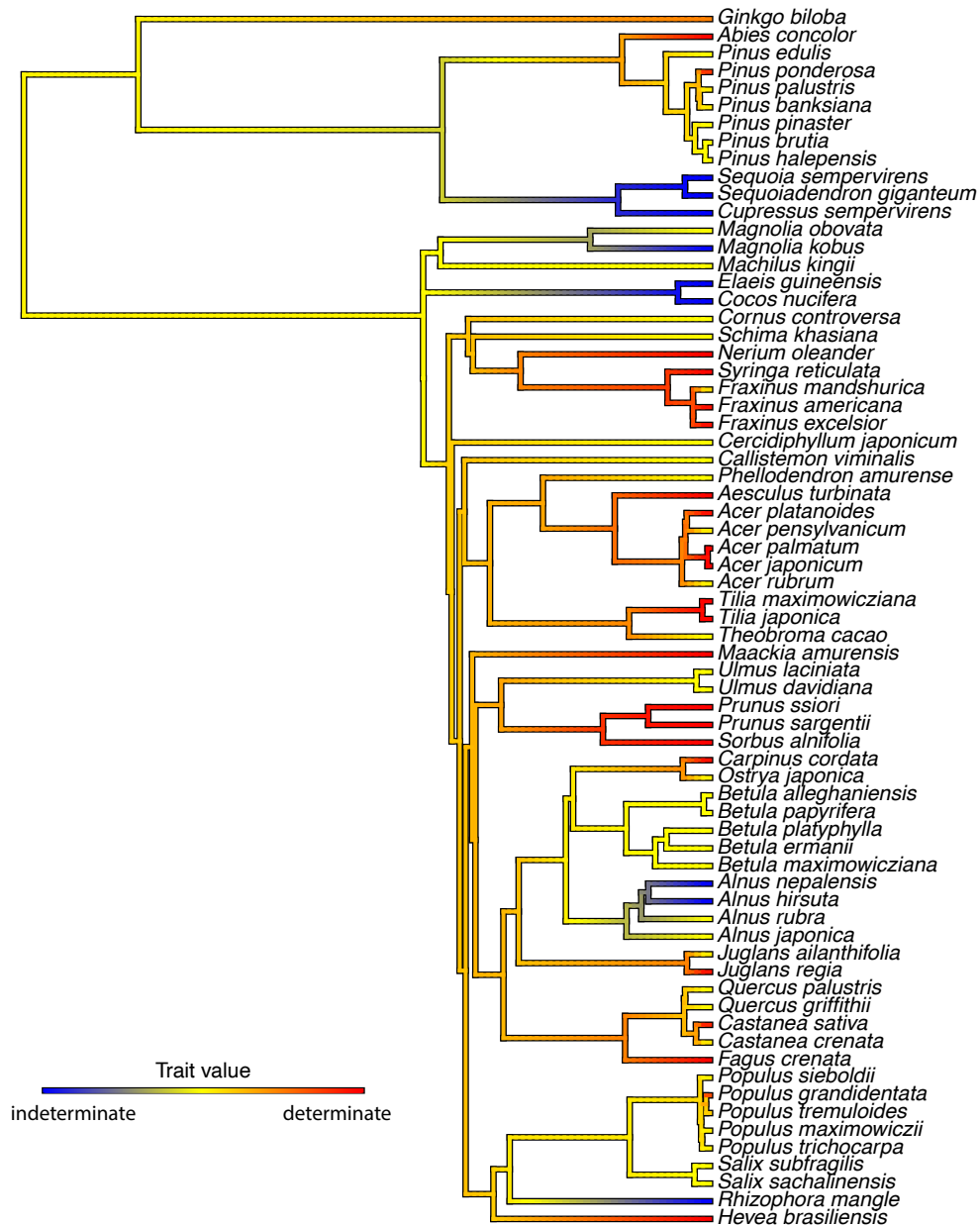


Figure 4: Phylogenetic tree of plant species colored by growth determinacy trait values, visually representing the evolutionary distribution of growth strategies across taxa. This tree includes species for which shoot growth determinacy data were compiled (mainly from Hallé *et al.* (1978) and Kikuzawa (1983)), with the phylogeny pruned from the comprehensive seed plant tree of Smith & Brown (2018) to match the trait dataset. Growth determinacy was scored as indeterminate (blue), intermediate (yellow) and determinate (red), with mean values calculated per species when multiple entries existed. These values were then mapped as a continuous trait onto the phylogeny using the `contMap()` function from the `phytools` R package.